

YASH KUMAR VERMA

AI ENGINEER | GenAI & Full-Stack Developer

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[Portfolio](#) | Available for Full-Time Roles Immediately | Open to Relocation

PROFESSIONAL SUMMARY

AI Engineer specializing in LLM systems, multi-agent RAG pipelines, model routing, and AI evaluation, with experience designing and deploying agentic applications end-to-end. Built agentic AI systems handling 50K+ daily queries at p99 <45ms, cutting LLM inference costs by 60% through dynamic routing across GPT-4, GPT-3.5, and fine-tuned Llama-3.1-8B. Strong grounding in Python, LangGraph, RAG evaluation, vector databases, and full-stack integration for shipping AI features end-to-end.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES2022+), TypeScript
- **AI/LLM Engineering:** LangChain, LangGraph, RAG, Prompt Engineering, Multi-Agent Systems, OpenAI API, Gemini API, LLM Evaluation (Ragas), Vector Databases (ChromaDB, Pinecone, Qdrant), Cross-Encoder Reranking, Embeddings, QLoRA, vLLM
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Pandas, NumPy, CNNs, Model Fine-Tuning
- **Backend & APIs:** Node.js, Express, FastAPI, REST APIs, Microservices, API Gateway, JWT Auth, Redis, Kafka, MySQL, MongoDB, PostgreSQL/PostGIS
- **Frontend & Mobile:** React, Next.js, TypeScript, React Native, Tailwind CSS, Vite, WebSocket, Monaco Editor
- **DevOps & Tools:** Docker, Git, AWS (EC2/Lambda), GCP, GitHub Actions, CI/CD, OpenTelemetry, LangSmith, Sentry, Vercel, Firebase

EXPERIENCE

Stealth AI Startup - AI Engineer Intern | Feb 2026 - May 2026

- Owned a 0-to-1 multi-agent RAG pipeline, rebuilding retrieval with hybrid search and cross-encoder reranking; scaled to 50K+ daily queries at p99 <45ms and improved Ragas faithfulness from 0.72 to 0.91.
- Cut LLM costs by 60% with an intent classifier routing queries across GPT-4, GPT-3.5, and fine-tuned Llama-3.1-8B using QLoRA/vLLM; reduced per-query cost from \$0.04 to \$0.016 while maintaining output quality.
- Shipped AI safety and observability with real-time PII scrubbing, prompt-injection defense, output moderation, and LangSmith tracing; blocked 99%+ adversarial inputs and improved task completion by 35% through A/B prompt testing.

Tellis Technologies - Software Development Intern | Sept 2025 - Jan 2026

- Built a multi-agent AI software engineer using LangGraph to localize GitHub issues, generate unified-diff patches, execute repository test suites, and iteratively repair failures through a Planner → Editor → Test Runner → Verifier loop.
- Implemented AST-level semantic retrieval with tree-sitter, extracting complete functions/classes and indexing them in ChromaDB with Gemini embeddings for issue-to-code matching.
- Engineered an ephemeral, network-isolated Docker sandbox with strict timeout enforcement for safe test execution, plus an automated SWE-bench-lite evaluation harness tracking pass/fail, iterations, and time-to-resolve.

Luminary - Software Developer Intern | Apr 2025 - Sept 2025

- Architected full-stack SaaS features serving 100K+ MAU: shipped micro-frontends with Module Federation, optimistic UI, WebSocket sync, and Node.js/FastAPI services behind an API Gateway, reducing Time-to-Interactive to 1.2s and bundle size by 55%.
- Redesigned an event-driven backend handling 10M+ daily events across 6 microservices using the Saga pattern, Kafka, circuit breakers, exponential backoff, and PostgreSQL read replicas with connection pooling; reduced p99 latency from 800ms to 35ms and achieved 99.95% uptime.
- Built GitHub Actions CI/CD with blue-green deployments, achieved 90%+ test coverage using Vitest and Playwright E2E, and added OpenTelemetry plus Sentry observability; cut deployment time from 30min to 8min and production incidents by 80%.

Elite Coders - Full Stack Development Intern | Feb 2025 - Mar 2025

- Developed core School ERP modules for student records, attendance, and fee management using the MERN stack for real institutional use.
- Optimized MongoDB schema design and indexing on high-traffic collections, improving query performance for ERP reporting workflows.
- Built and secured RESTful APIs with role-based access control for admin, teacher, and student roles, supporting multi-tenant institutional access.

AI PROJECTS

HyperGraph-CDC: Real-Time Event-Driven Knowledge Graph & MCP Agent Gateway | [GitHub](#)

PostgreSQL, Debezium, Kafka, Neo4j, Qdrant, LangGraph, MCP

- Built a real-time, event-driven data ingestion pipeline achieving <15ms database capture latency and a 1.1s end-to-end graph indexing turnaround by capturing PostgreSQL WAL changes via Debezium, streaming through Apache Kafka, and writing asynchronously to Neo4j and Qdrant.
- Developed a hybrid GraphRAG search engine combining dense vector search with Cypher-based k-hop graph traversals; used Gemini 2.5 Flash for entity/relationship extraction and maintained retrieval under 69ms while reducing grounding context blind spots.
- Architected an MCP supervisor agent gateway over Stdio transport with a LangGraph router to dynamically discover and route graph, vector, web-search, and codebase-indexing tools; validated sub-5ms Qdrant/Neo4j retrieval at 100K+ vectors, up to 25 concurrent write operations/sec, and exposed live telemetry through FastAPI + React/Vite.

Cortex AI — Multi-Agent AI Platform | [GitHub](#)

Node.js, React, LangGraph.js, Redis, MongoDB, Qdrant, Microservices

- Architected a microservices-based multi-agent platform with Auth, Chat, Agent, and Billing services behind an API Gateway, orchestrating 8 specialized agents through a LangGraph.js supervisor that auto-classifies intent and routes requests
- Built specialized agents for code generation (DeepSeek), web search (Tavily), PDF/PPT generation, AI image generation (Pollinations.ai), and multimodal vision (Gemini); implemented PDF RAG with Gemini embeddings and Qdrant Cloud, Redis conversation memory, and Firebase session auth.
- Designed API Gateway security and storage with Firebase Admin SDK auth, Redis sessions, Helmet headers, per-agent rate limiting, and Backblaze B2; instrumented LangSmith/MongoDB/Redis observability, LLM-as-a-judge evaluation, and cosine-similarity semantic caching to reduce duplicate-query latency and LLM cost.

Local Talk-to-Codebase RAG (CodeLens) | [GitHub](#)

LangChain, LangGraph, ChromaDB, Gemini, Ragas, Cross-Encoder Reranking

- Built a fully local retrieval-augmented generation system enabling natural-language Q&A over any codebase using LangGraph orchestration, ChromaDB vector storage, and Gemini for grounded, code-aware responses.
- Implemented Cross-Encoder reranking and a Ragas evaluation harness covering context precision, recall, faithfulness, and answer relevancy, reaching an internally validated 8/10 output-quality score.
- Hardened multi-tenant concurrent queries by resolving silent failure modes and stale reranker references; shipped a multi-stage Docker build with persistent ChromaDB volumes and GitHub Actions CI/CD running automated Ragas regression tests on every push.

Agent Observability Platform | [GitHub](#)

Python, FastAPI, PostgreSQL, React, Recharts, Docker, LangChain/LangGraph

- Built a plug-and-play Python instrumentation SDK for LangChain/LangGraph agents that streams traces to a FastAPI collector with asynchronous step-flush guarantees and zero blocking overhead.
- Engineered rule-based failure classification for timeouts (>30s), infinite loops, and hallucinations without external LLM calls; enforced try/finally database session lifecycle guarantees to eliminate connection leaks.
- Designed cross-version regression detection for success-rate drops >20% using chronological version timestamps and shipped a real-time React/Recharts dashboard with 3-second polling, latency/success-rate comparison, per-run cost analytics, and a LangGraph evaluation harness for healthy vs. degraded agents.

PUBLICATIONS & RESEARCH

- **Agentic RAG for Domain-Specific Knowledge Retrieval:** Co-authored a peer-reviewed academic paper, 2025.
- **Technical Report: SIH-25040/INCOIS:** Authored a 60+ page technical report for a Ministry of Earth Sciences-sponsored project on agentic retrieval systems, 2025.

EDUCATION

B.Tech in Computer Science & Engineering | Sep 2022 - Jun 2026

Vishveshwarya Group of Institutions, Greater Noida, UP | GPA: 7.87 / 10

LEADERSHIP & ACHIEVEMENTS

- **Solved 200+ LeetCode problems, demonstrating strong data structures, algorithms, and problem-solving practice.**
- **Top-3 finishes at 8+ inter-college hackathons, building and shipping functional AI/full-stack prototypes under time constraints.**